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Plant Biodiversity and Environmental Sustainability at Tunas Muda School

Service as Action — Final Research Report

RESEARCH GROUP

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Tunas Muda School Grounds · IB MYP Service as Action

1 Executive Summary

This research investigates plant biodiversity and environmental conditions within Tunas Muda School's grounds, aiming to identify opportunities to improve ecosystem health and encourage environmental responsibility among students.

Using photographic observations of trees, ornamental plants, landscaped areas and surrounding infrastructure, our group examined the visible diversity of plant life and its relationship with the school environment.

Our initial findings indicate that the school supports several types of vegetation, including flowering trees, palms, broad-leaved tropical plants and

ornamental shrubs. However, some plants exhibit visible signs of leaf discoloration and damage, while several planting areas are surrounded by paved surfaces and infrastructure.

> RESEARCH LIMITATION
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> Although these observations do not necessarily indicate poor ecosystem health, they highlight opportunities for further investigation, improved plant maintenance, biodiversity conservation and environmental education.

We propose developing a student-led biodiversity awareness website, introducing plant identification labels, conducting regular environmental monitoring and encouraging the school community to participate in conservation activities.

Through this project, we hope to demonstrate that environmental sustainability begins with understanding and appreciating the ecosystems within our immediate surroundings.

2 Introduction and Background

2.1 Background

Biodiversity refers to the variety of living organisms within an environment, including plants, animals, fungi and microorganisms. Plant biodiversity is particularly important because plants serve as primary producers, provide habitats, support food webs and contribute to environmental stability.

Within a school environment, plants also provide shade, improve the appearance of outdoor spaces and create opportunities for students to learn

about ecological relationships.

At Tunas Muda School, we noticed that the surrounding vegetation is sometimes overlooked in everyday school activities. This observation inspired our group to investigate the school's plant biodiversity and explore how students could become more environmentally responsible.

Rather than simply identifying plants, our project aims to understand their ecological significance, evaluate observable environmental conditions and recommend practical improvements.

2.2 Research Question

How does plant biodiversity contribute to the environmental health of Tunas Muda School, and what practical actions can the school community take to preserve and improve its ecosystem?

2.3 Research Objectives

01 Identify and document different types of plants found within the school grounds.

02 Observe visible indicators of plant health and environmental conditions.

03 Analyze the ecological importance of the observed vegetation.

04 Identify potential environmental challenges and areas for improvement.

05 Develop practical recommendations to increase environmental awareness and sustainability.

06 Communicate our findings through an accessible biodiversity research website and a structured environmental improvement plan.

3 Research Methodology

3.1 Research Approach

This project uses a qualitative observational approach, supported by photographic documentation.

Four photographs were provided as initial evidence of the school's vegetation. These photographs represent several outdoor environments, including a flowering tree near recreational facilities, ornamental planting beds, landscaped pathways and an area containing palms.

3.2 Data Collection

The photographic evidence was analyzed according to the following categories:

OBSERVATION CATEGORY	INDICATORS
Plant diversity	Different plant forms, leaf structures and growth patterns
Plant condition	Leaf color, visible damage, flowering and general appearance
Environmental conditions	Soil exposure, shade, nearby pavement and infrastructure
Ecological significance	Potential habitat, shade provision and biodiversity support
Human influence	Landscaping, planting containers, recreational facilities and pathways

3.3 Research Limitations

RESEARCH LIMITATION

The research is based on four photographs rather than a comprehensive ecological survey. Consequently, the findings cannot establish the total number of plant species, population sizes, soil quality, wildlife diversity or long-term ecosystem health.

RESEARCH LIMITATION

Plant identification is also preliminary, as accurate species identification may require closer photographs of leaves, flowers, fruits and stems.

RESEARCH LIMITATION

No soil tests, biodiversity indices, interviews or repeated monitoring results have been provided. These would strengthen the research if conducted in future sessions.

4 Research Findings and Observations

4.1 Observation 1 — Flowering Tree Near Recreational Facilities

◇ Image 1 — Flowering tree near recreational facilities. **Open record →**

The first photograph shows a mature, branching tree with numerous rounded, bilobed leaves and scattered pink-purple flowers. These characteristics suggest that the tree belongs to the genus Bauhinia, commonly known as the orchid tree.

PRELIMINARY IDENTIFICATION

Preliminary identification: the genus Bauhinia, suggested by visible characteristics. Species is not confirmed from the photograph.

Visible characteristics:

- A relatively large canopy with multiple branches.
- Green, rounded leaves with a characteristic divided shape.
- Scattered pink-purple flowers.
- Some yellowing leaves among predominantly green foliage.
- Surrounding recreational infrastructure, including seating areas and paved surfaces.

Ecological significance:

Flowering trees can provide nectar and other resources for visiting insects, while their branches and foliage may offer shelter for small organisms. Their canopies also provide shade and can reduce the amount of direct sunlight reaching the ground.

Environmental assessment:

ECOLOGICAL INTERPRETATION

The tree appears established and retains substantial foliage. However, some leaves are yellow, and the photograph does not provide sufficient evidence to determine whether this reflects normal leaf aging, seasonal changes or environmental stress.

RECOMMENDED ACTION

Recommended action: Monitor leaf condition over time, document flowering periods and investigate whether pollinating insects visit the flowers.

4.2 Observation 2 – Ornamental Planting Beds

Image 2 — Ornamental tropical plants in planting beds. **Open record →**

The second photograph presents several cultivated tropical plants growing in rectangular white planters and a larger planting bed. The visible plants include variegated ornamental foliage, deeply lobed leaves and large arrow-shaped leaves.

Preliminary plant identification:

- Aglaonema species or cultivars — likely represented by the variegated plants with green, cream and pinkish leaf patterns.
- Alocasia or a related aroid genus — possibly represented by the large, upright, arrow-shaped leaves.

- Thaumatophyllum or a related genus — potentially represented by the deeply lobed foliage.

Visible characteristics:

- Several different leaf shapes, sizes and color patterns.
- Organized planting containers.
- Exposed soil surrounding plant bases.
- Some yellowing leaves, brown leaf edges and localized damage.
- Plants growing close together in a landscaped environment.

Ecological significance:

These ornamental plants increase the visible variety of vegetation within the school. Their foliage provides additional plant cover and contributes to the complexity of the landscaped environment. However, the presence of different leaf colors and shapes does not automatically indicate high species diversity, as some plants may be cultivars of the same species.

Environmental assessment:

> ECOLOGICAL INTERPRETATION
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> The planting beds contain mostly green foliage, suggesting that many plants remain actively growing. Some individual leaves show discoloration or damage, but their causes cannot be determined from photographs alone.

> RECOMMENDED ACTION
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> Recommended action: Inspect irrigation and drainage, remove fully dead foliage when appropriate, and monitor whether leaf damage increases over time.

4.3 Observation 3 — Landscaped School Pathway

Image 3 — Landscaped school pathway and vegetation. **Open record →**

The third photograph shows a landscaped school environment containing a flowering tree, trimmed shrubs, a pedestrian pathway, outdoor seating and recreational infrastructure. The flowering tree appears similar to the one documented in Image 1 and may represent the same individual or species.

Visible characteristics:

- A flowering tree with an extensive canopy.
- Multiple maintained shrubs or small trees.
- Exposed soil surrounding some planted areas.

- Paved walkways and seating facilities.
- Vegetation distributed alongside human activity spaces.

Ecological significance:

The combination of trees and shrubs provides multiple vegetation layers. Different plant heights and structures can create a wider range of potential microhabitats than a landscape containing only grass. Trees also offer shade for students and help make outdoor spaces more comfortable.

Environmental assessment:

ECOLOGICAL INTERPRETATION

The vegetation is visibly maintained and integrated into the school's outdoor infrastructure. However, the photograph alone cannot establish whether the landscaping supports substantial wildlife diversity or whether the existing species are predominantly native or introduced.

RECOMMENDED ACTION

Recommended action: Establish a biodiversity observation zone where students can record plant species, flowering activity and visible wildlife without disturbing the habitat.

4.4 Observation 4 – Tropical Palms Near a Wall

EVIDENCE GAP

Listed as primary evidence; the image file was not available when this site was built.

◇ Image 4 — Tropical palms near a school boundary wall. Listed as primary evidence in this report; the image file was not available when the website was built. **Open record →**

The fourth photograph shows several palms growing alongside a boundary wall, with additional vegetation nearby. The palms have elongated, feather-like leaves consisting of multiple narrow leaflets.

PRELIMINARY IDENTIFICATION

Preliminary identification: these plants belong to the palm family, Arecaceae. They resemble commonly cultivated ornamental palms, although the exact species cannot be confirmed from the photograph.

Visible characteristics:

- Tall, slender stems.
- Long, arching, feather-like leaves.
- Dense foliage at different heights.
- Vegetation growing near a wall.
- Limited visible space between some plants.

Ecological significance:

Palms contribute to vegetation structure and may provide shelter or resources for small organisms, depending on their species and local ecological interactions. Their leaves can also contribute organic matter to the soil as they naturally age and decompose.

Environmental assessment:

ECOLOGICAL INTERPRETATION

The palms appear predominantly green and upright. The photograph does not reveal sufficient information about soil conditions, root development or long-term plant health.

RECOMMENDED ACTION

Recommended action: Monitor plant growth, inspect the surrounding soil and ensure maintenance practices are appropriate for the identified palm species.

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5 Overall Biodiversity Analysis

5.1 Diversity of Plant Forms

The photographic evidence demonstrates that Tunas Muda School contains multiple forms of vegetation, including:

- Mature flowering trees.
- Ornamental shrubs and small trees.
- Broad-leaved tropical plants.
- Variegated foliage plants.
- Palms and other landscaped vegetation.

> RESEARCH LIMITATION
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> This suggests that the school possesses a visually varied planted environment. However, because the photographs cover only selected locations, they cannot be used to calculate total species richness or establish the overall biodiversity of the school.

5.2 Plant Health and Environmental Conditions

Most photographed plants retain substantial green foliage, indicating the presence of living, established vegetation. Nevertheless, several ornamental plants show visible discoloration or leaf damage.

> ECOLOGICAL INTERPRETATION
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> Possible explanations include natural aging, water imbalance, environmental exposure, nutrient availability or pests. Further examination would be required before identifying any particular cause.

The presence of paved surfaces near planting areas also highlights the importance of investigating soil permeability, root space and water drainage.

5.3 Human Influence on the Ecosystem

The photographs show that vegetation is closely integrated with school facilities, including pathways, seating areas, recreational spaces and walls. This relationship creates opportunities for environmental education because students encounter plants throughout their daily routines.

However, landscaped ecosystems also require management. Plant selection, watering practices, pruning and waste management can influence their long-term ecological value.

> RESEARCH LIMITATION
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> The available evidence does not establish that students have damaged or neglected the plants. Student awareness and behavior should therefore be investigated separately rather than assumed.

6 Environmental Challenges and Opportunities

POTENTIAL CHALLENGE	SUPPORTING OBSERVATION	IMPROVEMENT OPPORTUNITY
Limited student awareness	Identified as the group's initial concern, not yet measured	Conduct awareness surveys and educational campaigns
Leaf discoloration and damage	Visible in Images 1 and 2	Introduce regular plant health monitoring
Unverified plant identities	Multiple plant types visible	Create a verified plant identification inventory
Unknown wildlife diversity	Wildlife not clearly documented in photographs	Conduct non-invasive wildlife observations
Limited ecological data	Only four photographs currently available	Establish repeated monitoring and standardized records
Human infrastructure near plants	Paths, walls and seating visible	Review soil conditions, drainage and available planting space

RESEARCH LIMITATION

These are research priorities rather than confirmed cases of environmental damage.

7 Proposed Environmental Improvements

Recommendation 1 — Create a School Biodiversity Website. Develop an accessible website documenting the plants found within Tunas Muda School. The website should include verified plant names, photographs, ecological roles, observation locations and practical conservation information. This would transform the group's findings into an educational resource available to students and teachers.

Recommendation 2 — Introduce Plant Identification Labels. Install small, weather-resistant labels near selected plants, subject to school approval. Each label could display the plant's common name, scientific name and a QR code linking to its profile on the biodiversity website. This would encourage students to recognize and appreciate the vegetation they encounter.

Recommendation 3 — Establish Regular Plant Health Monitoring. Create a student-led monitoring program that records changes in leaf condition, flowering, visible pests and environmental conditions. Monitoring should be carried out consistently so that observations can be compared over time. Potential concerns should be communicated to school maintenance staff rather than treated through unapproved interventions.

Recommendation 4 — Encourage Ecologically Appropriate Planting. Work with school staff to evaluate whether suitable native or locally beneficial plants could be introduced into appropriate planting areas. Species should be selected based on their ecological value, compatibility with the local environment, maintenance requirements and available space. This could help support local biodiversity while preserving the functionality of school facilities.

Recommendation 5 — Organize Environmental Awareness Activities. Conduct short educational campaigns through school announcements, posters, digital media or student presentations. Activities could encourage students to recognize plant species, respect landscaped areas and participate in non-invasive biodiversity observations.

RECOMMENDED ACTION

Each recommendation is expanded on this website with its evidence, implementation steps, dependencies and evaluation criteria: see Improvements.

8 Proposed Implementation Plan

PHASE	ACTION	INTENDED OUTCOME
1 — Documentation	Collect additional photographs and verify plant identities	Reliable biodiversity inventory
2 — Research	Investigate ecological roles and environmental conditions	Stronger scientific understanding
3 — Digital Awareness	Publish the biodiversity research website	Accessible student learning resource
4 — Coordination	Share findings and recommendations with appropriate school staff	Feedback and implementation guidance
5 — Implementation	Introduce approved labels, monitoring or awareness activities	Greater student participation
6 — Evaluation	Compare observations and collect feedback	Evidence of progress and areas for improvement

RESEARCH LIMITATION

The timeline should be finalized after consultation with the school.

9 Connection to the Sustainable Development Goals

SDG 15 — Life on Land

This project contributes to SDG 15 by encouraging the protection, observation and responsible management of terrestrial biodiversity. Through identifying plant species and investigating their ecological significance, students can develop a stronger understanding of the importance of healthy ecosystems.

SDG 11 — Sustainable Cities and Communities

School green spaces form part of the wider urban environment. Maintaining these spaces can support more sustainable, comfortable and environmentally responsible communities.

SDG 4 — Quality Education

The proposed biodiversity website and awareness activities provide opportunities for environmental education, scientific inquiry and student-led learning.

> RESEARCH LIMITATION
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> These connections describe how the project relates to the goals. The project does not claim to have achieved measurable outcomes against any SDG target.

10 Service as Action Reflection

This project encourages us to recognize that environmental responsibility extends beyond large-scale issues such as climate change and deforestation. It also involves appreciating and protecting the living organisms within our immediate surroundings.

By examining the vegetation at Tunas Muda School, we developed a clearer understanding of how plants contribute to the environment and how scientific observation can reveal opportunities for improvement.

Our research also highlighted the importance of distinguishing evidence from assumptions. Although photographs can reveal visible plant characteristics, they cannot independently explain the causes of plant damage or establish the overall health of an ecosystem.

As a group, we aim to apply our research through environmental education, collaborative planning and realistic recommendations to school leadership.

Our intended contribution is not simply to produce a report, but to encourage a lasting sense of responsibility toward the school's natural environment.

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ll Conclusion

The photographic observations reveal a range of trees, palms and ornamental plants across Tunas Muda School's outdoor environment. These plants contribute to the school's vegetation structure, provide shade and offer potential ecological resources for other organisms.

Although some plants display visible leaf discoloration or damage, the available evidence is insufficient to determine whether these observations indicate significant environmental problems.

Our findings therefore emphasize the need for continued investigation, verified plant identification and regular environmental monitoring.

By developing a biodiversity website, promoting environmental awareness and proposing practical conservation measures, our group hopes to encourage students to recognize the ecological importance of their surroundings.

Ultimately, a more sustainable school environment depends not only on the presence of plants, but also on the knowledge, cooperation and responsibility of the community that shares the space.

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12 Environmental Improvement Recommendations and Action Plan

12.1 Overview

Through our Service as Action research, we have documented several forms of vegetation within the school's grounds and identified opportunities to strengthen environmental awareness and biodiversity monitoring. This section gathers those opportunities into a practical set of environmental improvement recommendations for the school community.

12.2 Supporting Research Observations

The recommendations below rest on the four photographic observations documented in this report: a flowering tree near the recreational facilities, ornamental planting beds, a landscaped pathway and tropical palms near a boundary wall. Visible leaf discoloration in Observations 1 and 2, unverified plant identities across the grounds and limited ecological data frame both the opportunities and the limitations of what is proposed.

12.3 Proposed Interventions

- Digital biodiversity documentation — an accessible website recording observed plants, their characteristics and ecological significance.
- Plant identification and education — verified identification labels, potentially supported by QR codes linking to individual database entries.
- Plant health monitoring — regular observations of plant condition, flowering activity and visible environmental changes.
- Biodiversity-friendly planting — investigation of suitable native or locally beneficial plants that could enhance appropriate school areas.
- Environmental awareness — activities that encourage students to recognize, appreciate and responsibly interact with the school's vegetation.

12.4 Implementation Requirements

Implementation depends on coordination with appropriate school staff. Placement of identification labels, changes to planting areas and scheduled monitoring activities should be agreed with the school before any physical changes are made. Potential plant health concerns should be communicated to school maintenance staff rather than treated through unapproved interventions.

12.5 Expected Educational and Ecological Benefits

These measures could help students better understand the importance of the school's natural environment while encouraging responsible behavior and long-term sustainability. Environmental education would be embedded in students' daily routes, and evidence-based care would replace assumption with observation.

12.6 Monitoring and Evaluation

Progress could be evaluated through the number of verified observation records maintained on the website, the consistency of completed monitoring sessions, the number of identification labels installed where permitted, participation in awareness activities and documented changes in vegetation composition over time.

12.7 Long-Term Sustainability

The initiative is designed to be maintained by students, with this website serving as a durable record that future groups can extend. Long-term success depends not only on the presence of plants, but on the knowledge, cooperation and responsibility of the community that shares the space.

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> RECOMMENDED ACTION
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> A dedicated, printable version of this action plan is available on the Improvement Plan page.

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13 Evidence and References

Primary Evidence

- Photograph 1 — Flowering tree near school recreational facilities.
- Photograph 2 — Ornamental tropical plants in school planting beds.
- Photograph 3 — Landscaped school pathway and vegetation.
- Photograph 4 — Tropical palms near a school boundary wall.

OBSERVED EVIDENCE

Evidence source: Photographs supplied by the research group, October 2026 (exact capture dates not specified). Photograph 4 was listed as evidence in this report but its file was not available when the website was built; its record is kept open until it is found.

Plant Identification Dossier Sources

The group's supplementary plant dossier cites the following sources, consulted for the Bauhinia and Araceae reference profiles: Flona Jakarta (flona.jakarta.go.id); Droen Solo Baru (droensolobaru.com); UConn Home & Garden Education Center (homegarden.cahnr.uconn.edu).

Suggested Authoritative Reference Sources

- Convention on Biological Diversity — https://www.cbd.int/
- Royal Botanic Gardens, Kew: Plants of the World Online — https://powo.science.kew.org/
- United Nations — Sustainable Development Goal 15: https://sdgs.un.org/goals/goal15
- United Nations — Sustainable Development Goal 11: https://sdgs.un.org/goals/goal11
- United Nations — Sustainable Development Goal 4: https://sdgs.un.org/goals/goal4

RESEARCH LIMITATION

These are starting points for verifying scientific information and preparing a complete Harvard-style reference list. They have not been presented as sources already consulted by the group.

End of report. This digital edition preserves the complete text of the original document, including its limitations. Photograph 4 remains listed as primary evidence and will be added when its file is found.